

Figure 1: Network monitoring ensures all-round security for networks using firewalls (Image credits: www.paessler.com)

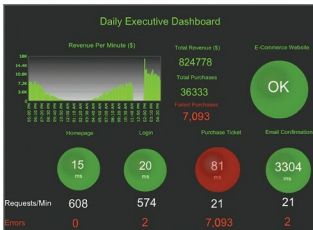


Figure 2: Sample analysis report of the Hypervic tool (Image credits: <http://www.computerworlduk.com/>)

order to ensure that the virus scanners and different anti-malware programs on each computer within the company are up-to-date. This ensures that the client computers are continuously protected against any malware as well.

3. Network monitoring solutions help the administrator measure the bandwidth for leased lines, devices (routers, switches), network connections, etc. Detailed monitoring of the bandwidth usage can also indirectly detect malware attacks. An indication of such an attack may be slow response times from different applications and websites, caused by a malware program that actually eats up a large amount of the bandwidth.

A few open source network monitoring tools

Here are some of the open source network monitoring tools widely available in the market.

Hypervic: This has originated at VMware. It has been developed for monitoring different custom Web applications and their performance across all physical, virtual and cloud environments. Hypervic works across Web servers, application servers, databases, operating systems, messaging servers, hypervisors and directory servers. This network monitoring tool offers an enterprise version that helps in improving the alerting functions and is also able to create better baselines.

Functionality and usage

It helps in:

- OS monitoring
- Detailed reporting
- Application and middleware monitoring
- Alerts and remediation workflows

Zenoss Core: This is another open source stalwart which gives network administrators a complete solution for tracking and managing various applications, servers, networking components, storage, virtualisation tools, etc. Administrators can make sure that the hardware is running efficiently and they can even take advantage of the modular design to plug in different ZenPacks for extended functionality. Zenoss Core 5 was released in February 2016 in order to improve the already powerful tool with an enhanced user interface and an expanded dashboard. It's Web-based console and dashboards were already highly dynamic and customisable. The new version now helps administrators mash up various components' charts onto a single chart. Hence, it's actually the tool for better root cause analysis.

Functionality and usage

It helps in:

- Network mapping
- Monitoring device issues, daemon processes and production states by listing different event views
- Out-of-band management and monitoring of all Zenoss components
- Online backup, restore, snapshots and multi-host deployment

Xymon: This is a significant network monitoring tool which was formerly known as Hobbit. It was developed to address the shortcomings of tools like Big Brother and Big Sister. It's actually very easy to deploy Xymon on any system and it is, of course, available free of cost.

Functionality and usage

It helps in:

- Monitoring servers, applications and networks
- Offering information about the health of the various components networked via Web pages



Figure 3: Status report of the Xymon tool (Image credits: <http://www.computerworlduk.com/>)